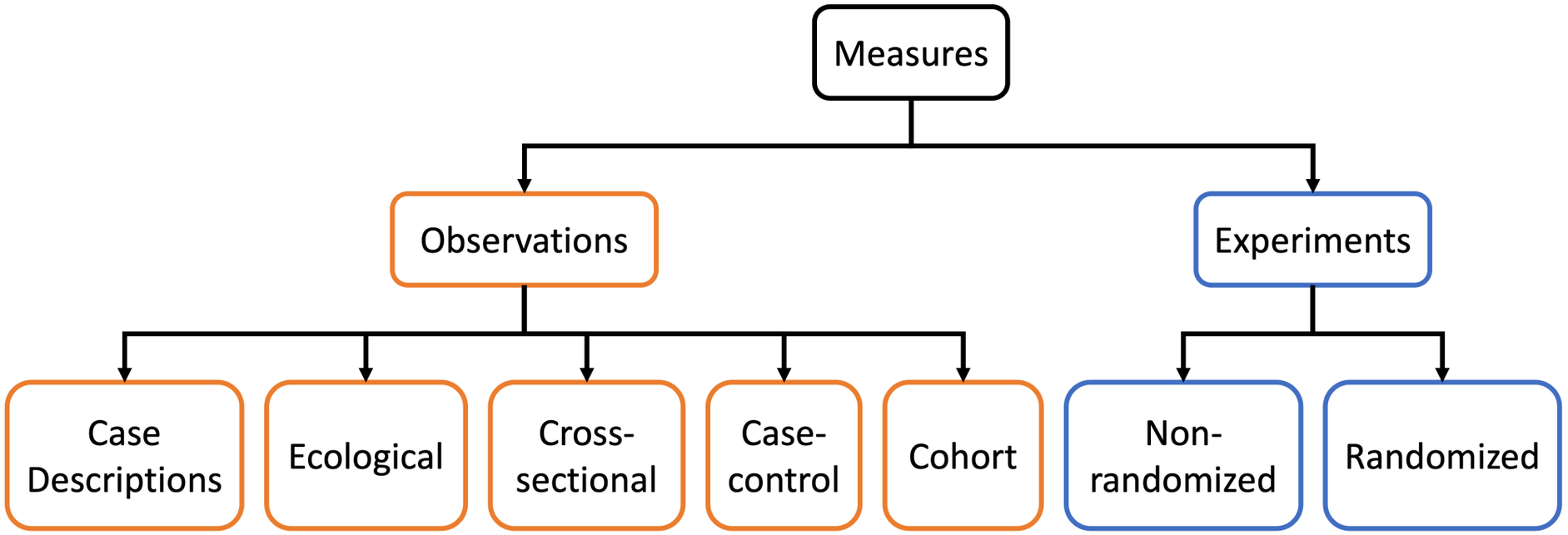
Case-Control Studies

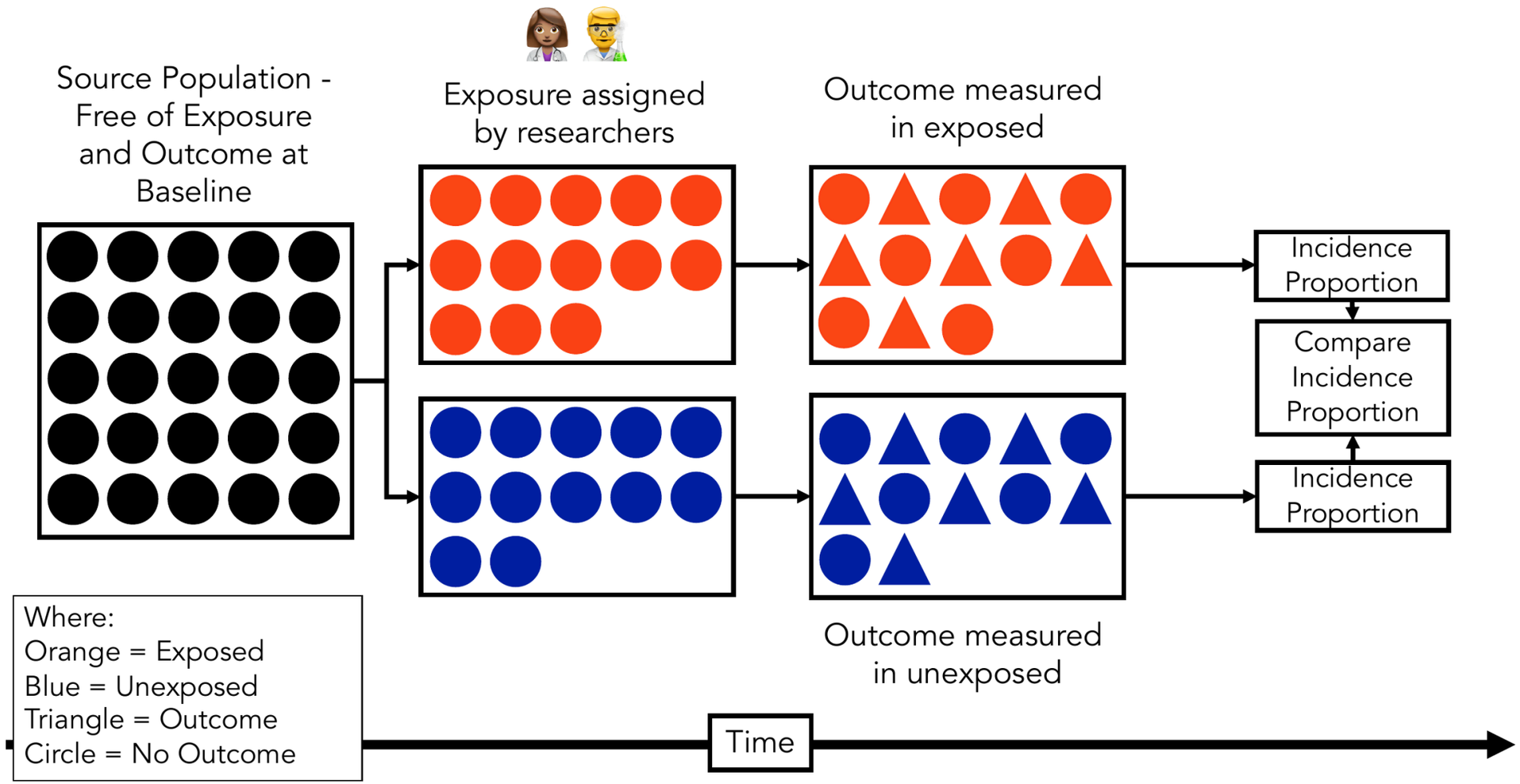
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Hierarchy of Study Designs

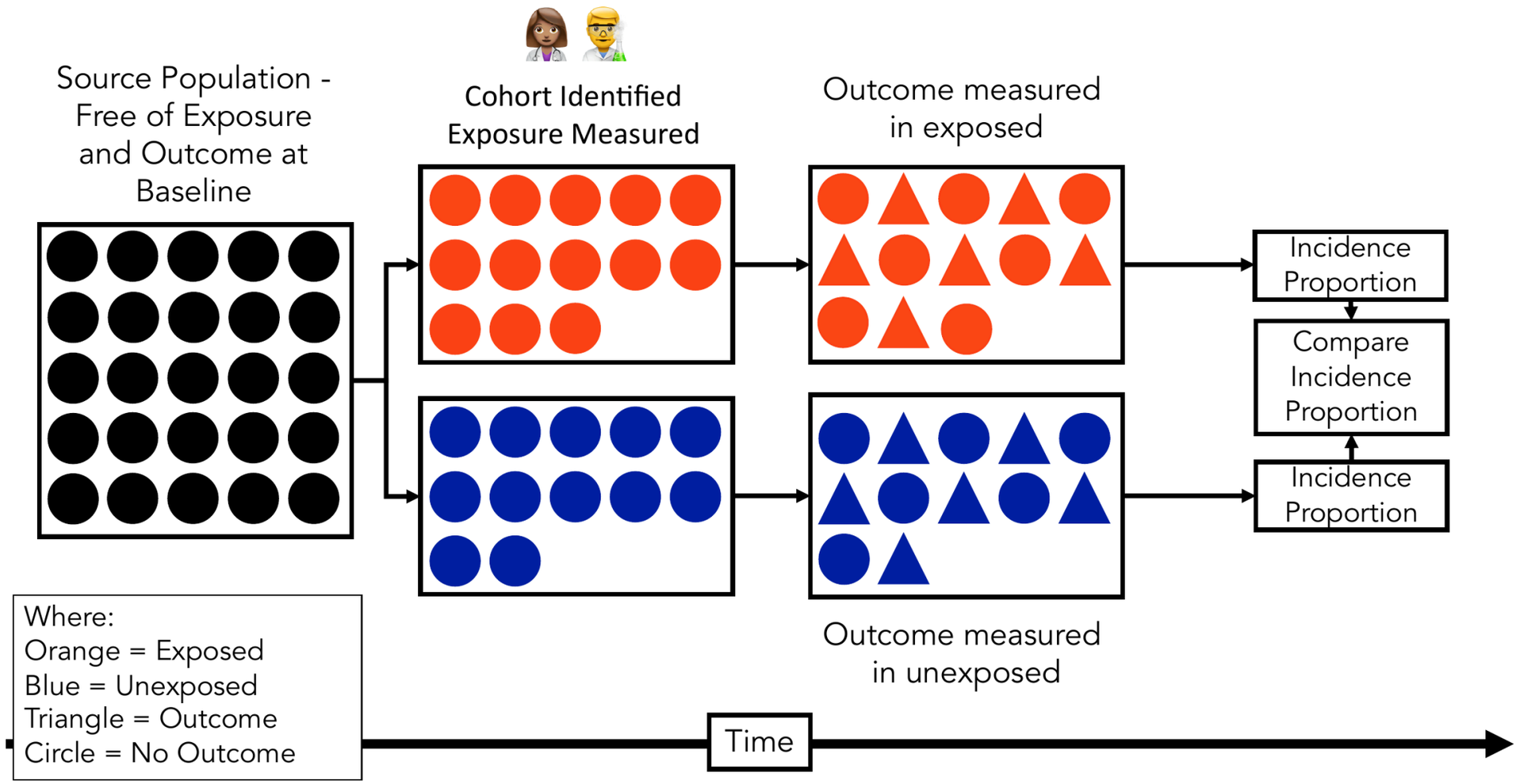


Typically less convincing → Typically more convincing

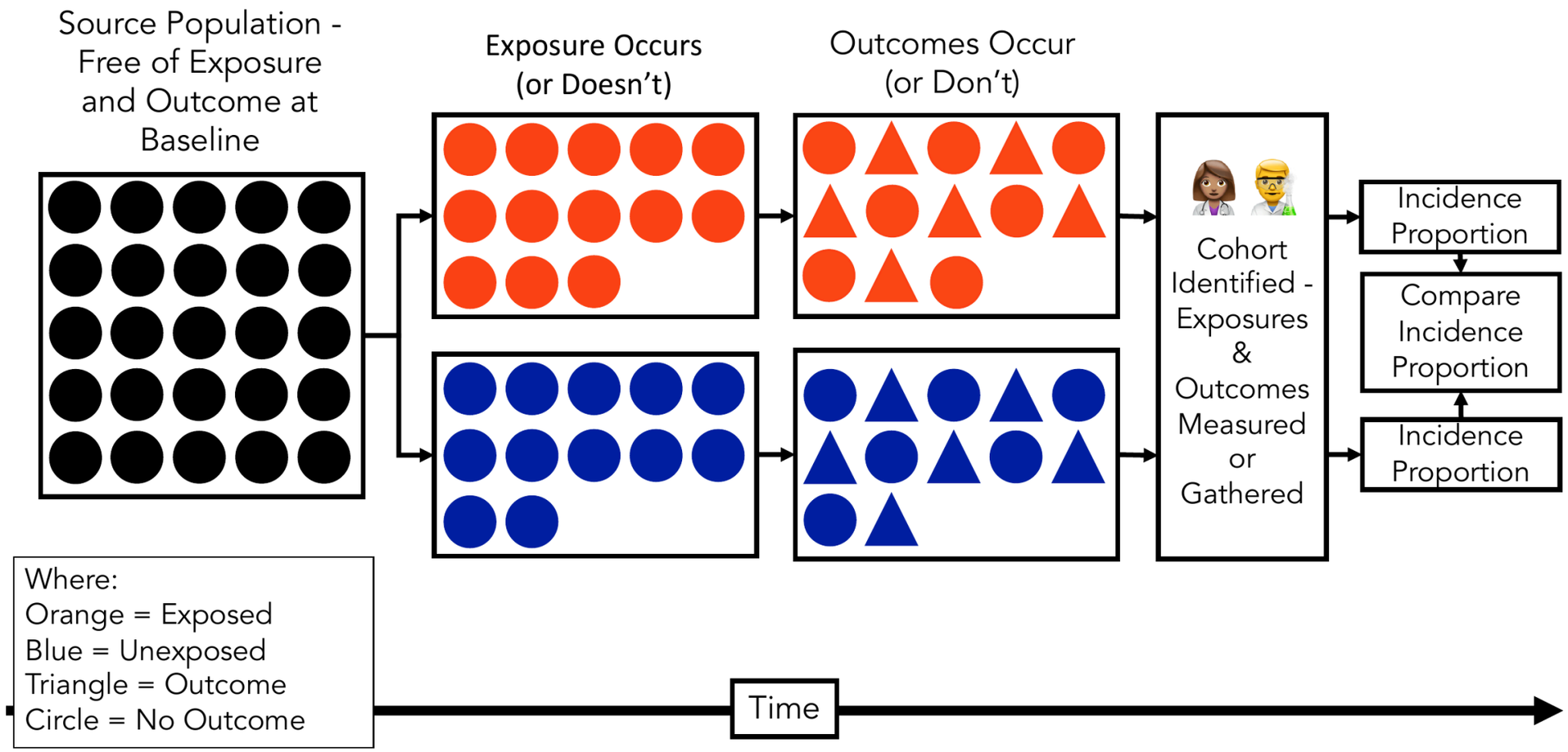
Basic Experimental Study Design



Common Prospective Cohort Study Design



Common Retrospective Cohort Study Design

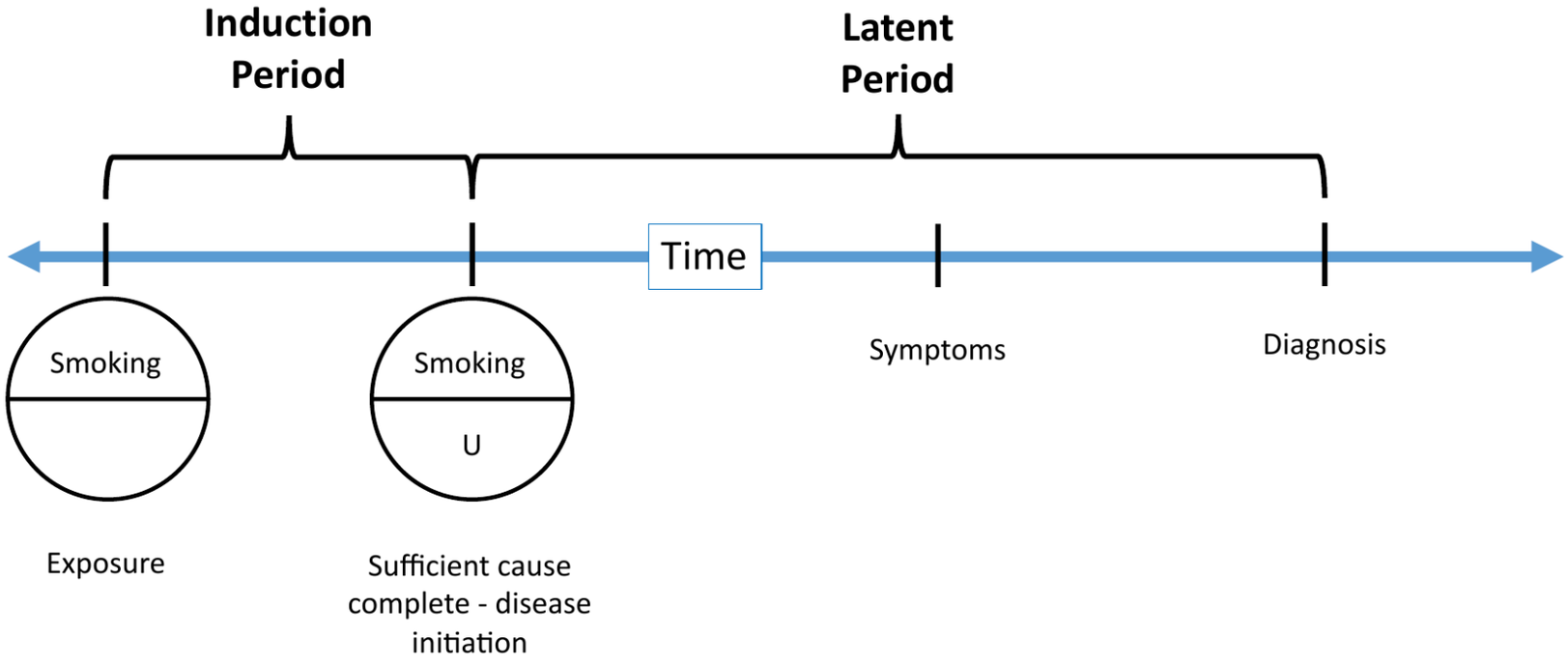


When Cohort Is Not Ideal

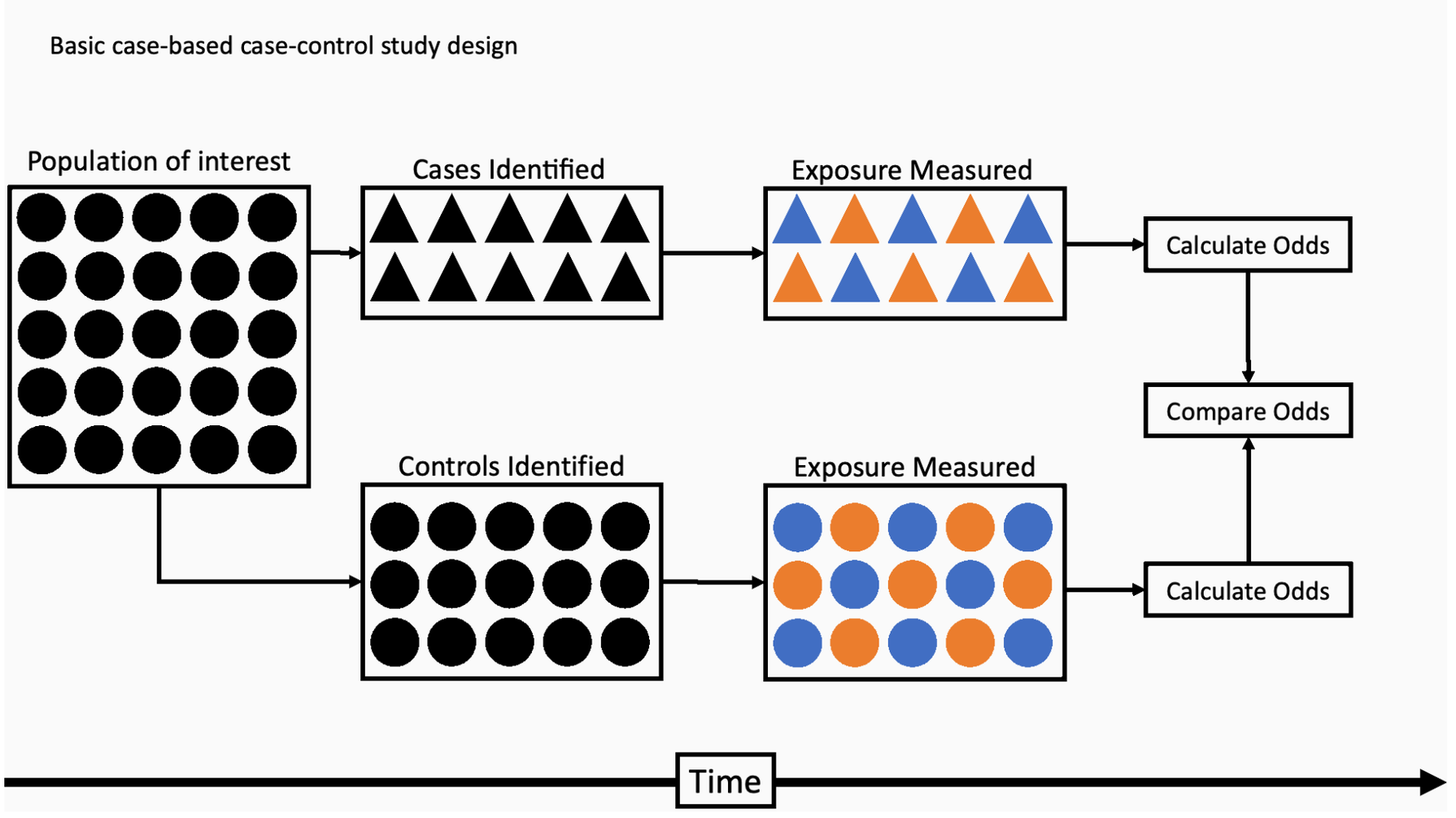
Cohort studies are powerful — but not always practical.

- What if:
  - The disease is **rare**?
  - The outcome has a **long latency period**?
  - Long-term follow-up is **too costly or infeasible**?
  - You need results **quickly**?
  - In these situations, a different design may be more efficient.

Induction and Latent Periods



Basic Case-Control Study Design



Retrospective Cohort vs. Case-Control

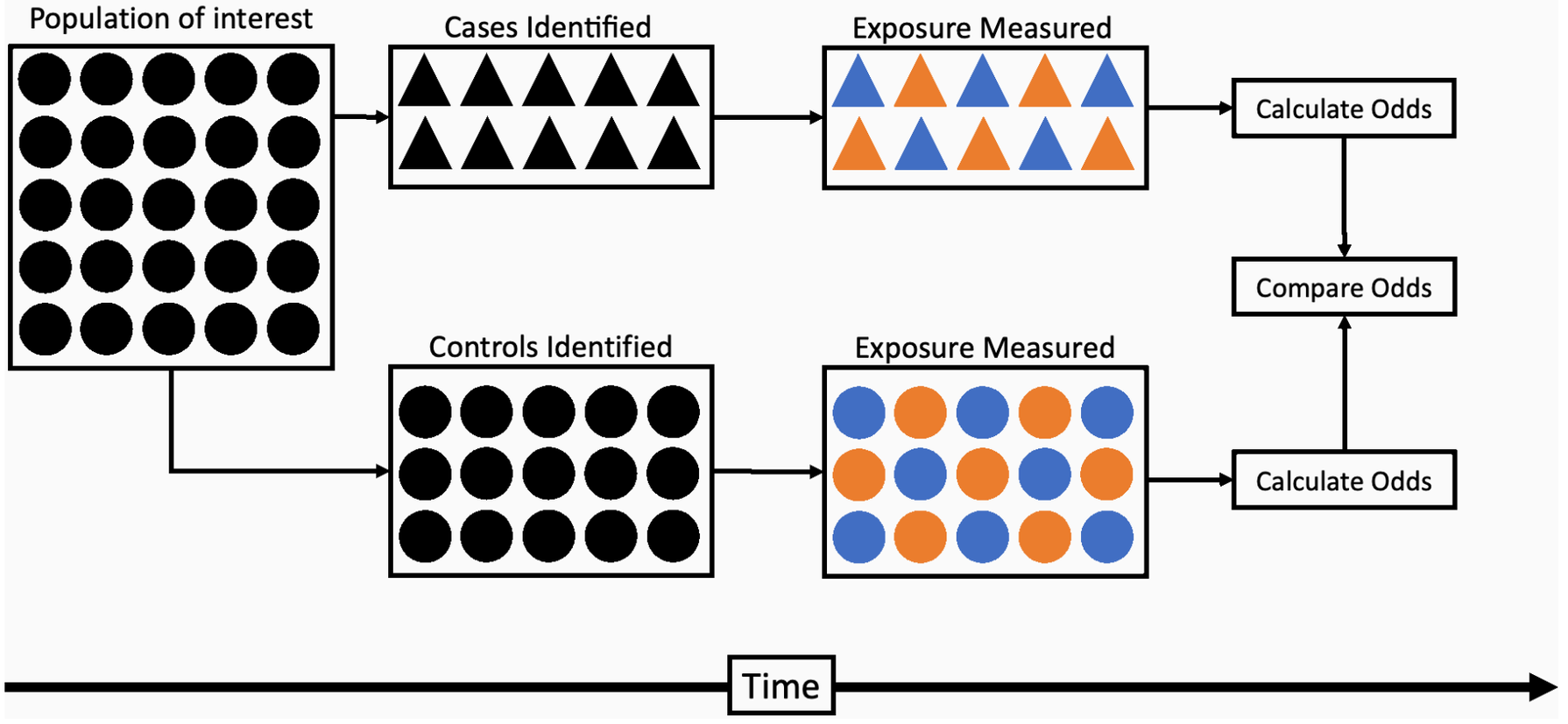
- Case-control
  - Sample based on outcome and look backward for exposure.
- Retrospective Cohort
  - Sample based on exposure and look “forward” towards the outcome
  - A cohort study was done. You just weren’t there to initially observe it

Threats to Validity

Study Base Principle

- Cases and controls should be “representative of the same base experience.”
- If a potential control had gotten the disease, they would have been selected with the same probability as any other case.

Basic case-based case-control study design



Recall Bias

- **Definition:** A type of information bias in which participants remember past exposures **differently**.
- **When it occurs**
  - Often in **case-control studies**
  - Cases may recall exposures more carefully than controls
- **Key idea:** Differences in memory between groups can distort the exposure–outcome association.



Interviewer and Observer Bias

- **Interviewer Bias**
  - Occurs during **data collection**
  - Interviewer knowledge influences how questions are asked
  - Can lead to different responses between groups
- **Observer Bias**
  - Occurs during **outcome assessment**
  - Observer expectations influence how outcomes are recorded
  - More likely when blinding is absent
- **Key idea:** When researchers influence data collection or measurement, results may be systematically distorted.

Why Use an Odds Ratio?

- In case-control studies, we cannot calculate **risk** directly. Why?
  - We start with a fixed number of **cases and controls**
  - The total population at risk is unknown
  - Incidence cannot be measured
  - Instead, we quantify association using the **Odds Ratio (OR)**
- **What Does OR Measure?**
  - The **odds of prior exposure** among cases compared to the odds of exposure among controls
- **Key Contrast:**
  - Cohort → Uses **Relative Risk (RR)**
  - Case-Control → Uses **Odds Ratio (OR)**

Analysis of Case-Control Studies

Odds of Exposure Among Cases

Given the prototypical 2x2 table for case-control data, which of the formulas below will calculate the odds of exposure among cases?

	Cases	Controls
Exposed	A	B
Not exposed	C	D

A. $\frac{a}{c}$

B. $\frac{a}{b}$

C. $\frac{a}{a+b}$

D. $\frac{a}{a+c}$

Odds of Exposure Among Cases

Given the prototypical 2x2 table for case-control data, which of the following is an accurate expression of the odds of exposure among cases?

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	???	

- The odds of exposure among cases = $a/c = 50/100 = 0.5$.
- Equivalent ways to express these odds: **50:100 = 0.5:1 = 5:10 = 1:2**.
- For every **1 exposed case**, there are **2 unexposed cases**.

Odds of Exposure Among Controls

Given the prototypical 2x2 table for case-control data, which of the following is an accurate expression of the odds of exposure among controls?

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	???

A. 0.2

B. 1:2

C. 2.5:1

D. 0.25

Odds of Exposure

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

- Odds of exposure among cases = 50 / 100 = **0.5**
- Odds of exposure among controls = 25 / 100 = **0.25**

Association

- The distribution of the thing we are measuring differs, on average, between two groups. This could be its center, spread, shape, or the proportion of people in each category.
- Knowing something about **X** tells us something about **Y**.

Comparing Exposure Odds

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

- Odds of exposure among cases = **0.5**
- Odds of exposure among controls = **0.25**

The odds are higher among cases. **How much higher?**

Calculating the Exposure Odds Ratio

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

$$\text{Exposure OR} = \frac{\text{Odds of exposure among cases}}{\text{Odds of exposure among controls}} = \frac{a/c}{b/d}$$

Odds of Exposure Among Controls

Given the prototypical 2x2 table for case-control data, what is the exposure odds ratio?

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

A. 1

B. 2

C. 3

D. 4

Exposure Odds Ratio: Calculation

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

$$\text{Exposure OR} = \frac{a/c}{b/d} = \frac{50/100}{25/100} = \frac{0.5}{0.25} = 2$$

How would you interpret this odds ratio?

Exposure Odds Ratio: Interpretation

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

$$\text{Exposure OR} = \frac{50/100}{25/100} = \frac{0.5}{0.25} = 2$$

The odds of exposure among cases are **2 times** the odds of exposure among controls.

Why Use the Odds Ratio?

	Cases	Controls
Exposed	50	25
Not exposed	100	100
Odds of exposure	50 / 100	25 / 100

- We can calculate it directly from a case-control study.
- We can also estimate it using logistic regression.
- Under certain conditions, it approximates the **risk ratio**.

Hypothetical Cohort: Hypertension and MI

One-year incidence of acute myocardial infarction (**MI**, or heart attack). Severe systolic hypertension: ≥180 mm Hg. Normal systolic blood pressure: <120 mm Hg.

	MI (+)	MI (–)	Total
Severe hypertension	180	9,820	10,000
Normotensive	30	9,970	10,000

- Risk of MI among people with severe hypertension = 180 / 10,000 = **0.018**
- Risk of MI among normotensive people = 30 / 10,000 = **0.003**
- Risk ratio (incidence proportion ratio) = 0.018 / 0.003 = **6**
- Participants with severe hypertension had **6 times the risk of MI** over one year compared with normotensive participants.

Risk and Odds

	Outcome	No outcome	Total
Exposed	a	b	a + b
Unexposed	c	d	c + d

Measure	Formula	a = 0	Incidence = 0.001
Incidence	$\frac{a}{a+b}$	$\frac{0}{0+b} = 0$	0.001
Odds	$\frac{\frac{a}{a+b}}{1-\frac{a}{a+b}}$	$\frac{\frac{0}{0+b}}{1-\frac{0}{0+b}} = \frac{0}{1-0} = \frac{0}{1} = 0$	$\frac{0.001}{1-0.001} \approx \frac{0.001}{1}$

Risk in the exposed: $p = a/(a + b)$. Odds in the exposed: $p/(1 - p) = a/b$.

When risk is small, **1 – risk is close to 1**, so odds and risk are similar.

Comparing the Risk Ratio and Odds Ratio

	MI (+)	MI (–)	Total
Severe hypertension	180	9,820	10,000
Normotensive	30	9,970	10,000

- The incidence of MI among people with severe hypertension = 180 / 10,000 = **0.018**
- The incidence of MI among people who are normotensive = 30 / 10,000 = **0.003**
- **Risk ratio = 0.018 / 0.003 = 6**
- The odds of severe hypertension among cases = 180 / 30 = **6**
- The odds of severe hypertension among controls = 9,820 / 9,970 ≈ **0.985**
- **Exposure OR = (180 / 30) / (9,820 / 9,970) ≈ 6.09**

Hypertension and MI: Interpreting the OR

	MI (+)	MI (–)	Total
Severe hypertension	180	9,820	10,000
Normotensive	30	9,970	10,000

- Exposure odds among cases = 180 / 30 = **6**
- Exposure odds among non-cases = 9,820 / 9,970 ≈ **0.985**
- **Exposure OR = (180 / 30) / (9,820 / 9,970) ≈ 6.09**

Participants with MI had **6.09 times the odds of severe hypertension** compared with participants who did not have an MI during the one-year study period.

Sampling 10% of Controls

	MI (+)	MI (–)	Cohort total
Severe hypertension	180	982	??
Normotensive	30	997	??

- Population risks require the original cohort denominators: **180 / ??** and **30 / ??**.
- Exposure odds among cases = 180 / 30 = **6**
- Exposure odds among sampled controls = 982 / 997 ≈ **0.985**
- **Exposure OR = (180 / 30) / (982 / 997) ≈ 6.09**

The OR is unchanged when we sample the same fraction of exposed and unexposed non-cases.

Disease Odds Ratio

	MI (+)	MI (–)	Cohort total
Severe hypertension	180	9,820	??
Normotensive	30	9,970	??

- Disease odds with severe hypertension = 180 / 9,820 ≈ **0.01833**
- Disease odds with normal blood pressure = 30 / 9,970 ≈ **0.003009**
- **Disease OR = (180 / 9,820) / (30 / 9,970) ≈ 6.09**

Participants with severe hypertension had **6.09 times the odds of MI** over one year compared with normotensive participants.

Disease OR With 10% of Controls

	MI (+)	MI (–)	Cohort total
Severe hypertension	180	982	??
Normotensive	30	997	??

- Sample odds with severe hypertension = 180 / 982 ≈ **0.1833**
- Sample odds with normal blood pressure = 30 / 997 ≈ **0.03009**
- **Disease OR = (180 / 982) / (30 / 997) ≈ 6.09**

Participants with severe hypertension had **6.09 times the odds of MI** over one year compared with normotensive participants. The individual sample odds differ from the population odds.

Interpreting the Odds Ratio

- **Interpreting the Odds Ratio (OR)**
  - **OR = 1:** No association between exposure and disease
  - **OR > 1:** Exposure is associated with **higher odds** of disease
  - **OR < 1:** Exposure is associated with **lower odds** of disease

Strengths and Limitations

Strengths

- Efficient for studying **rare diseases**
- Faster and less costly than cohort studies
- Require **fewer participants**
- Useful for diseases with **long latency**

Limitations

- Cannot directly measure **incidence or risk**
- Vulnerable to **selection bias**
- Prone to **recall bias**
- Control selection can be challenging

Cohort vs. Case-Control Studies

Feature	Cohort Study	Case-Control Study
Start With	Exposure	Outcome
Direction	Exposure → Outcome	Outcome → Exposure
Measure	Relative Risk (RR)	Odds Ratio (OR)
Best For	Rare exposures	Rare diseases
Time & Cost	Longer, more expensive	Faster, less expensive
Main Bias Concern	Loss to follow-up	Selection & recall bias
Can Measure Incidence?	Yes	No

Practice: Sampling

In a case-control study, we typically sample based on the _____ we are interested in studying.

A. Exposure

B. Outcome

Practice: Identify the Study Design

Researchers survey 1,000 adults at a single point in time, asking about their current smoking status and whether they currently have a chronic cough. What type of study is this?

A. Cohort study

B. Case-control study

C. Cross-sectional study

D. Randomized trial

Practice: Identify the Study Design

Researchers use medical records from 2010 to identify 1,000 smokers and 1,000 nonsmokers who were free of lung cancer. They review records through 2020 to determine who developed lung cancer. All records already exist when the study begins. What type of study is this?

A. Cohort study

B. Case-control study

C. Cross-sectional study

D. Randomized trial

Practice: Identify the Study Design

Researchers sample 500 patients who have lung cancer and 500 patients who do not have lung cancer. They compare the two groups' smoking histories before the lung cancer diagnoses. What type of study is this?

A. Cohort study

B. Case-control study

C. Cross-sectional study

D. Randomized trial

Practice: Study Base Principle

Shands Hospital at the University of Florida is a referral hospital for a variety of cancer diagnoses -- including bronchogenic carcinoma -- for the North Florida region where it is located.

Researchers are interested in the association between bronchogenic carcinoma and a history of cigarette smoking, and they decide to conduct a case-control study. Cases are selected from the admissions records to the cancer center. Controls are randomly selected from the population of the surrounding county.

Does the control group violate the study base principle?

A. Probably

B. Probably not

Practice: Control Selection

Researchers are interested in the effect of obesity on colorectal cancer and conduct a case-control study. Cases are selected from a large colon cancer registry. Researchers are concerned that unmeasured genetic factors may bias study results.

What would be a good control group to remove the potential confounding effects of genetic factors?

Practice: Identifying Bias

In a case-control study investigating the relationship between alcohol use and liver cancer, cases were recruited from a major urban hospital. Controls were recruited from a local fitness club.

Which bias is most likely?

A. Recall bias

B. Attrition bias

C. Selection bias

D. Interviewer bias

Practice: Interpreting Odds

In a study of Parkinson's disease, the odds of prior pesticide exposure among cases were **0.25**, or **1 to 4**.

What does this mean?

A. 25% of cases had prior pesticide exposure

B. For every 1 case with prior pesticide exposure, there were 4 cases without prior exposure

C. Cases had 25% greater risk of pesticide exposure

D. Cases were 4 times as likely to have prior pesticide exposure as to have no prior exposure

Practice: Interpreting an Odds Ratio

In a case-control study, researchers compared patients with Parkinson's disease to controls without Parkinson's disease. The odds ratio for prior pesticide exposure was **2.5**.

What does this mean?

A. Pesticide exposure causes Parkinson's disease

B. People exposed to pesticides had 2.5 times the risk of Parkinson's disease

C. Parkinson's disease cases had 2.5 times the odds of prior pesticide exposure compared to controls

D. 2.5% of Parkinson's disease cases had prior pesticide exposure

Practice: Interpreting an Odds Ratio

In a case-control study, researchers compared patients with Alzheimer's disease to controls without Alzheimer's disease. The odds ratio for regular physical activity during midlife was **0.60**.

What does this mean?

A. Alzheimer's disease cases had 40% lower odds of regular physical activity during midlife compared to controls

B. Physically active people had 60% lower risk of Alzheimer's disease

C. Regular physical activity prevents Alzheimer's disease

D. There was no association between physical activity and Alzheimer's disease